

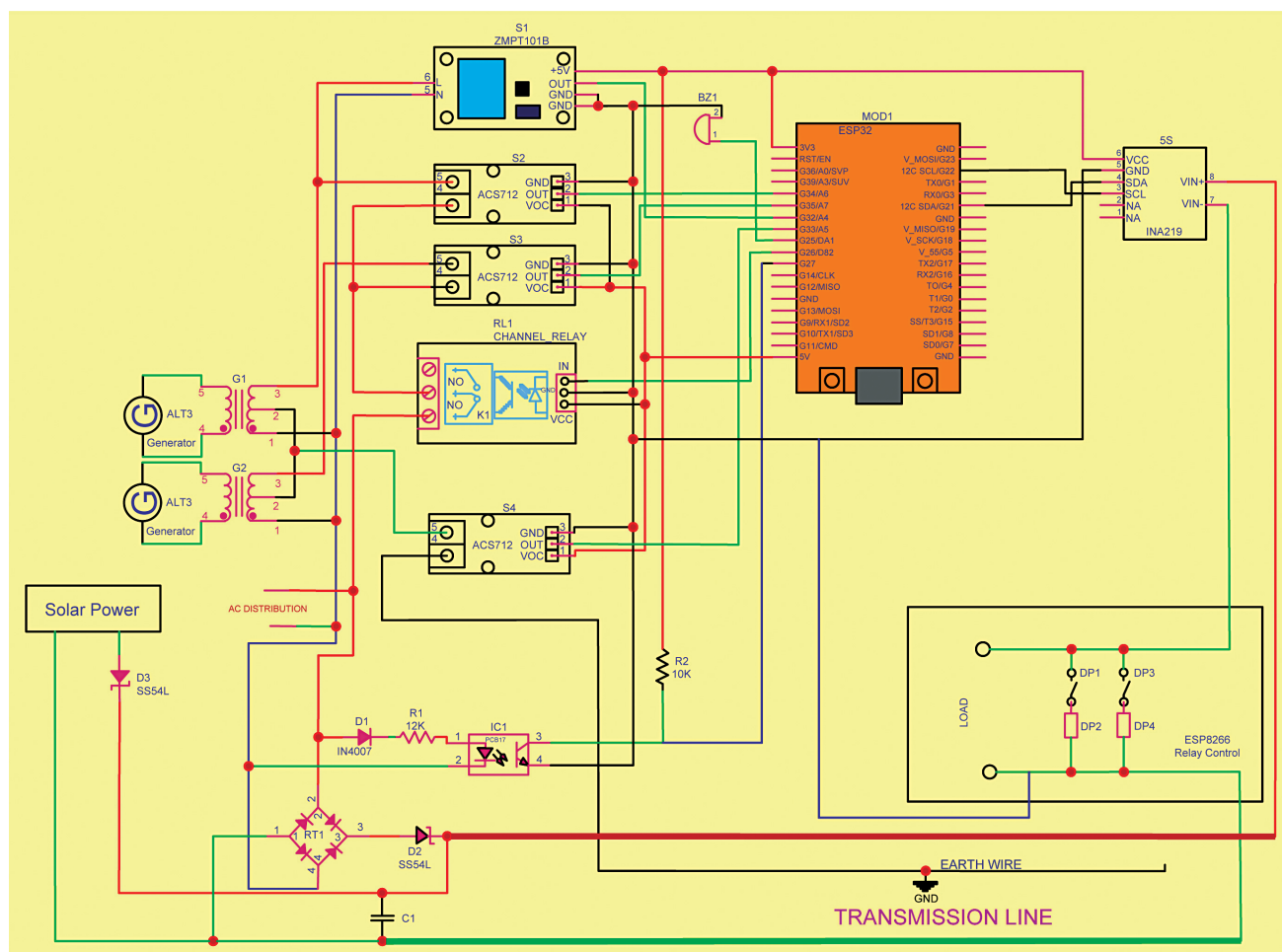


Fig. 3: Generator-side circuit

modules: one for the generator side and one for the load side. The load side uses the ESP8266 and a relay

module, as shown in Fig. 1. The generator side uses the ESP32 along with voltage and current sensors, as shown in Fig. 2. The components required to assemble the system are listed in the Bill of Materials table.

Design

The entire system is divided into three main sections.

1. Transmission line section. This section manages both AC and DC power sources along with their respective loads. For safety, the 230V AC supply is stepped down to 18V AC using two transformers. The current from each generating station is measured using ACS712 sensors (G1 and G2), while earth-fault current is

monitored using another ACS712 sensor. AC voltage on the generation side is measured through a ZMPT101B sensor, and grid frequency is detected using an optocoupler (PC718) interfaced with an ESP32 microcontroller.

The AC power is converted to DC using a rectifier and fed into the DC transmission line, where voltage and current are monitored using an INA219 sensor. As solar power increases, the current drawn from the AC source decreases, reducing fuel injection in the thermal generator and improving energy efficiency.

The DC voltage derived from the AC supply is kept slightly lower than the solar panel's output voltage, enabling automatic switching between sources. During daytime, when solar voltage is higher, the diode on the solar side becomes

BILL OF MATERIALS

Components	Name	Quantity
ESP32 WROVER	MOD1	1
ESP 8266	MOD2	1
INA219 current and voltage sensor	S5	1
Buzzer	BZ1	1
ZMPT101B voltage sensor	S1	1
ACS712 current sensor	S2, S3, S4	3
Single channel relay 5V	RL1	2
PC817 optocoupler	IC1	1
1N4007 diode	D1	1
230V AC/9-0-9V, 4A transformer	G1, G2	2
SS54L diode	D3, D4	2
12KΩ resistor	R1	1
10KΩ resistor	R2	1
1000μF, 25V electrolytic capacitor	C1	1
Bridge rectifier (KBL06)	RT1	1